

Exercise

- 1) Find the e-values and e-vectors of the matrix

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 11 \end{bmatrix}$$

Since A is triangular it has e-values of $\lambda = 11, 4, 1$.

$$\lambda_1 = 11: \quad A - 11I = \begin{bmatrix} -10 & -2 & 3 \\ 0 & -7 & 5 \\ 0 & 0 & 0 \end{bmatrix}$$

Now eliminating,

$$\begin{bmatrix} 1 & 1/5 & -3/10 \\ 0 & 1 & -5/7 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{-\frac{1}{5}r_2} \begin{bmatrix} 1 & 0 & -11/70 \\ 0 & 1 & -5/7 \\ 0 & 0 & 0 \end{bmatrix}$$

and setting $(A - 11I)\vec{v}_1 = \vec{0}$ we have

$$v_{11} = \frac{11}{70} v_{13}$$

$$v_{12} = \frac{5}{7} v_{13}$$

and thus $\vec{v}_1 = \begin{bmatrix} 11/70 \\ 5/7 \\ 1 \end{bmatrix}$

$$\lambda_2 = 4: \quad A - 4I = \begin{bmatrix} -3 & -2 & 3 \\ 0 & 0 & 5 \\ 0 & 0 & 7 \end{bmatrix}$$

Setting $(A - 4I)\vec{v}_2 = \vec{0}$ where $v_{23} = 0$

we have that $-3v_{21} - 2v_{22} = 0$

and thus $\vec{v}_2 = \begin{bmatrix} -2/3 \\ 1 \\ 0 \end{bmatrix}$

$$\lambda_1 = 1: \quad A - I = \begin{bmatrix} 0 & -2 & 3 \\ 0 & 3 & 5 \\ 0 & 0 & 10 \end{bmatrix}$$

$$\vec{v}_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Fundamental Theorem of Algebra: If

$p = \sum_{i=0}^n a_i x^i$, then p has exactly n roots in $\mathbb{C}$

(counting roots) and p has unique factorization given by $p(x) = c(x-x_1)^{n_1}(x-x_2)^{n_2}\dots(x-x_k)^{n_k}$ where n_i is the multiplicity of the zero $x=x_i$.
(order)

Def 14.1 The order (multiplicity) of a zero of the characteristic polynomial is called the algebraic multiplicity and dimension of the corresponding eigenspace is called the geometric multiplicity.

Example: If $p_A(\lambda) = (\lambda-3)^4 q(\lambda)$ and $\text{nullity}(A-3I) = 3$ we would say that the e-value $\lambda=3$ has algebraic mult. of 4 and geometric mult. of 3.

Theorem 14.2 Let λ_i be an e-value of A of order (mult.) n_i . Then

$$1 \leq \text{nullity}(A - \lambda_i I) \leq n_i$$

geometric mult. $\leq$ alg. mult.

"Proof" Assume that nullity $(A - \lambda_i I) = k$ and let $\{\vec{v}_1, \dots, \vec{v}_k\}$ be a basis for $\text{null}(A - \lambda_i I)$. Then extending to a basis $\mathcal{B}$ of $\mathbb{C}^n$ we have that

$$A\vec{v}_j = \lambda_i \vec{v}_j$$

$$[A]_{\mathcal{B}} = \begin{bmatrix} \lambda_i I_k & B \\ 0 & C \end{bmatrix}$$

$$\begin{aligned} \text{Now } |A - \lambda I| &= \begin{vmatrix} (\lambda_i - \lambda) I_k & B \\ 0 & C - \lambda I_{n-k} \end{vmatrix} \\ &= |(\lambda_i - \lambda) I_k| \cdot |C - \lambda I_{n-k}| \\ &= (\lambda_i - \lambda)^k |I_k| p_C \\ &= (\lambda_i - \lambda)^k p_C \end{aligned}$$

and thus the algebraic mult. of λ_i is at least k .

Polynomials and Annihilators

Def 14.3 A polynomial $p = \sum_{i=0}^n a_i x^i$ is monic if $a_n = 1$.

Def 14.4 Given polynomial $p = \sum_{i=0}^n a_i x^i$ and matrix $A \in \mathbb{C}^{n \times n}$ or $T \in \mathcal{L}(\mathbb{C}^n, \mathbb{C}^n)$ then $p(A) \in \mathbb{C}^{n \times n}$ and $p(T) \in \mathcal{L}(\mathbb{C}^n, \mathbb{C}^n)$ respectively where

matrix/
operator
polynomials

$$\left\{ \begin{aligned} p(A) &= \sum_{i=0}^n a_i A^i = a_0 I + a_1 A + a_2 A^2 + \dots \\ p(T) &= \sum_{i=0}^n a_i T^i = a_0 I + a_1 T + a_2 T^2 + \dots \end{aligned} \right.$$

$$\begin{aligned} p(A) \cdot \vec{x} &= (a_0 I + a_1 A + \dots + a_n A^n) \cdot \vec{x} \\ &= a_0 \vec{x} + a_1 A \cdot \vec{x} + \dots + a_n A^n \vec{x} \end{aligned}$$

Def 14.5 A polynomial p is an annihilator

of $A \in \mathbb{C}^{n \times n}$ ($T \in \mathcal{L}(\mathbb{C}^n, \mathbb{C}^n)$) if and only if $p(A)$ ($p(T)$) is the zero matrix. ($p(A) \cdot \vec{x} = \vec{0}$ for all $\vec{x} \in \mathbb{C}^n$)

Theorem 14.6 (Cayley-Hamilton) let

$T \in \mathcal{L}(\mathbb{C}^n, \mathbb{C}^n)$ and let $A = [T]_{\beta}$.

If $p(\lambda) = |A - \lambda I|$, then p is an annihilator of A and T . I.e.

$$p(A) \cdot \vec{x} = p(T)(\vec{x}) = \vec{0} \text{ for all } \vec{x} \in \mathbb{C}^n.$$

let $A \in \mathbb{C}^{n \times n}$ have linearly independent e-vectors of $\vec{v}_1, \dots, \vec{v}_n$ with corresponding e-values of $\lambda_1, \dots, \lambda_n$. Then

$$p_A(\lambda) = (\lambda_1 - \lambda)(\lambda_2 - \lambda) \cdots (\lambda_n - \lambda) \text{ and}$$

$$\text{thus } p_A(A) = (\lambda_1 I - A) \cdots (\lambda_n I - A)$$

Now let $\vec{x} \in \mathbb{C}^n$ be chosen arb. and

set $\vec{x} = \sum c_i \vec{v}_i$. Since each

$\vec{v}_i \in \text{null}(A - \lambda_i I)$ we have that

* This proof is for the special case of a diagonalizable matrix. The full proof follows from the Jordan form or from a density argument

$$p_A(A) \cdot \vec{x} = \sum c_i p_A(A) \cdot \vec{v}_i = \sum c_i \cdot \vec{0} = \vec{0}$$

as $(\lambda_j I - A)$ and $(\lambda_k I - A)$ commute.

$$\text{Symbolically: } p_A(A) = |A - AI| = 0$$

Minimal Polynomial

Def 14.7 The minimal polynomial of A is the minimal, monic annihilator of A .
 I.e. If μ_A is minimal polynomial and $p(A)$ is the zero matrix where p is monic, then $\deg \mu_A \leq \deg p$.

Notation: Given matrix A

- i. $p_A(\lambda) = |A - \lambda I|$ is the characteristic polynomial
- ii. $\mu_A(x)$ is the minimal polynomial

Examples:

- i. I has minimal polynomial of $\mu = x - 1$
- ii. If $P \neq I$ is a projection then $\mu_P = x^2 - x$

Theorem 14.8 The minimal polynomial divides the characteristic polynomial and if λ_0 is an e-value of A , then λ_0 is a zero of μ_A .

Given A :

- i. μ_A and p_A have the same zeros (possibly different mult.)
- ii. $p_A = \mu_A \cdot q$ where q is also a (monic) polynomial

Proof: Let λ_0 be an e-value of A with e-vector $\vec{v}_0 \neq \vec{0}$. Now let $f = \sum a_i x^i$

$$A \vec{v}_0 = \lambda_0 \vec{v}_0$$

$$\Downarrow$$

$$A^i \vec{v}_0 = \lambda_0^i \vec{v}_0$$

be any monic polynomial such that $f(A)$ is the zero matrix. Now since

$$f(\lambda_0) \vec{v}_0 = \sum a_i \lambda_0^i \vec{v}_0 = \sum a_i A^i \vec{v}_0$$

$$= f(A) \cdot \vec{v}_0$$

$$= \vec{0}$$

we have $f(\lambda_0) = 0$ as desired.

Finding the minimal polynomial: let $A \in \mathbb{C}^{n \times n}$ and pick basis $\{\vec{v}_1, \dots, \vec{v}_n\}$. Now for each i the set $\{\vec{v}_i, \underbrace{A\vec{v}_i, \dots, A^n \vec{v}_i}_{n+1}\}$ is linearly dep and set μ_i to be the minimal polynomial such that $\mu_i(A) \cdot \vec{v}_i = \vec{0}$. Then

$$\boxed{\mu_A = \text{LCM}(\mu_i)}$$

Exercise:

2) Find the minimal polynomials of the following

a. $A = \begin{bmatrix} 3 & 7 \\ -1 & 2 \end{bmatrix}$

$$\vec{e}_1: A\vec{e}_1 = \begin{bmatrix} 3 \\ -1 \end{bmatrix}, A^2\vec{e}_1 = \begin{bmatrix} 3 & 7 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \\ -5 \end{bmatrix}$$

Now as $\begin{bmatrix} 2 \\ -5 \end{bmatrix} = 5 \begin{bmatrix} 3 \\ -1 \end{bmatrix} - 13 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

we have that $A^2 \vec{e}_1 = 5A\vec{e}_1 - 13\vec{e}_1$,

or $(A^2 - 5A + 13I)\vec{e}_1 = \vec{0}$ and

thus $\mu_1 = x^2 - 5x + 13$. Now

as $\deg \mu_A \leq 2$, $\mu_A = x^2 - 5x + 13$.

You can verify that $(A^2 - 5A + 13I)\vec{e}_2 = \vec{0}$
as well.